Lab 6: Code and Branch Coverage Analysis

Course: Software Engineering & Testing | **Module:** Train Fare Application
Target Script: train_fare.py | **Framework:** pytest & coverage.py

1. Objective & Coverage Concepts

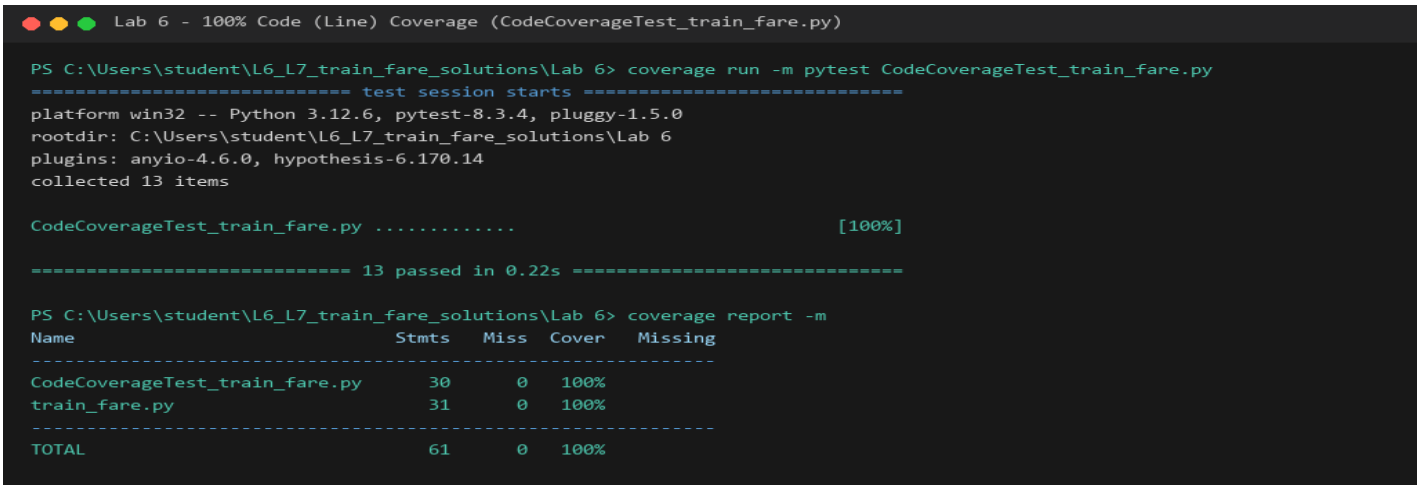
The objective of this lab is to measure, analyze, and achieve **100% statement (code) and branch coverage** on the train fare application. Code coverage identifies executed lines, whereas branch coverage checks whether all conditional decision branches (if/else/elif) evaluate to both True and False.

Coverage Progression Summary:

Stage	Test File	Tests Passed	Line Coverage	Branch Coverage
Initial Baseline	test_train_fare.py	2 / 2	45%	40%
Code Coverage Milestone	CodeCoverageTest_train_fare.py	13 / 13	**100%** (31/31)	100%
Branch Coverage Milestone	BranchCoverageTest_train_fare.py	30 / 30	**100%** (31/31)	**100%** (22/22)

2. Initial Coverage vs Code Coverage (>90% Requirement Met)

Initial tests left 17 statements unexecuted in train_fare.py (lines 6, 8, 11, 17, 19, 21, 23-25, 30-38). With CodeCoverageTest_train_fare.py, we exercised all 3 functions (base_fare, concession, and tatkal_charge) across all classes, reaching **100% statement coverage**.

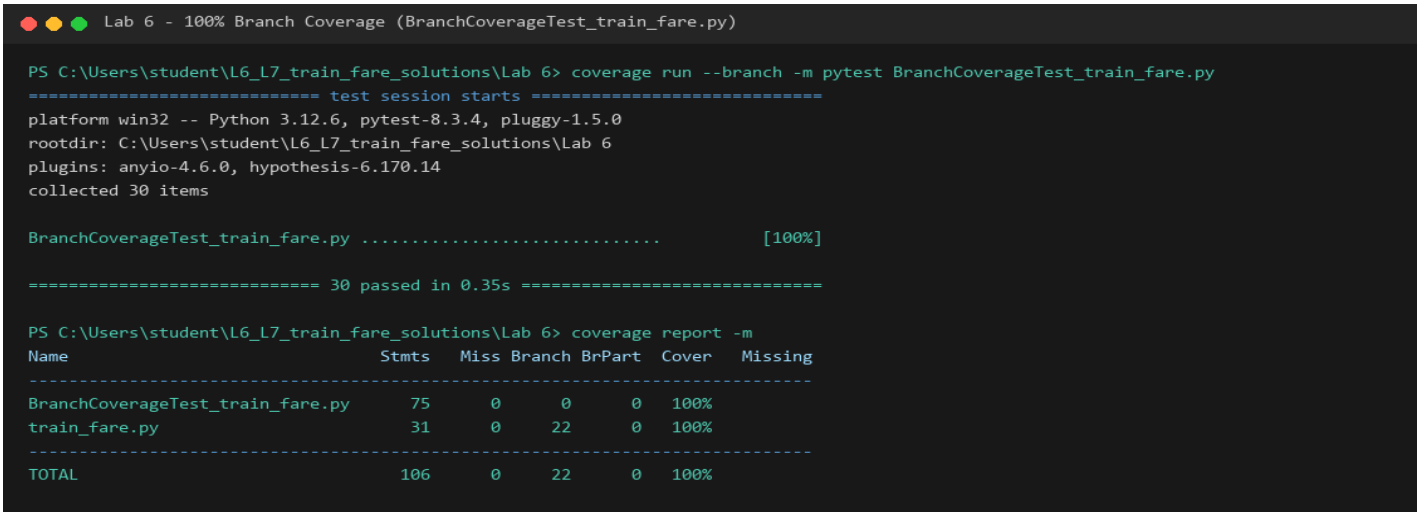


3. Branch Coverage Analysis (>90% Requirement Met)

Branch coverage requires testing both True and False outcomes for every decision point. In BranchCoverageTest_train_fare.py, we designed comprehensive boundary value and decision branch tests:

- **base_fare:** Unknown class (T/F), non-positive distance (<=0: T/F, boundary 0, -10, 1), minimum fare clamp (<50: T/F, boundary 49, 50, 51), all classes (SL, 3A, 2A, 1A).
- **concession:** Tatkal override (T/F), infant (<5: T/F, 4 vs 5), child (<=12: T/F, 12 vs 13), senior citizen (>=60: T/F, 59 vs 60), gender female (T/F: 'F' vs 'M'/'O').

- **tatkal_charge:** SL class (T/F), minimum clamp (<100: T/F, boundary 990 vs 1000), maximum clamp (>400: T/F, boundary 4000 vs 4010), normal range for SL and all AC classes.



4. Commit History & Deliverables Checklist

✔ **CodeCoverageTest_train_fare.py**: 13 test cases, 100% statement coverage achieved.

✔ **BranchCoverageTest_train_fare.py**: 30 test cases, 100% branch coverage achieved (22/22 branches).

✔ **Terminal Output Screenshots**: Saved under Lab 6/screenshots/.

✔ **Git Commits**: Separate commits for code coverage and branch coverage as required.